Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/08/2008
PI Name:	Liu, Yuanming Ph.D.		
Project Title:	He-4 Studies in a Low-gravity Simulator		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Bio-nanotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Jet Propulsion Laboratory		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8099	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	10/01/2002	**End Date:**	09/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Project extended to 9/30/2007 per D. Strayer, JPL (3/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	none		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The task is a ground-based, scientific research experiment, which will study the effects of a heat current on the helium-4 superfluid transition in a low-gravity simulator. This system represents a class of condensed matter physical systems that are driven out of equilibrium and exhibit critical dynamic behaviors near the phase transitions. This area of research has recently attracted more theoretical and experimental attentions that have yielded exciting new developments and new understandings. However, the agreements between experiments and theories are mostly qualitative; large discrepancies still exist. It is widely believed that gravity in earth-bound experiments may play a significant role in creating those discrepancies. Using the low-gravity simulator we have developed at JPL, we will be able to study the critical dynamic properties of helium and the superfluid transition more closely to the transition temperature in a reduced gravity environment. Also we		

	will be able to investigate the intrinsic effects of gravity on the physical properties under study and the artificial influences on the measurements themselves. We believe the results of the proposed experiment will benefit the low temperature helium physics community at large and the fundamental microgravity physics community in particular.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will demonstrate the research benefit of low-gravity simulator that utilizes magnetic levitation to counter gravity on a body of fluid.
Task Progress:	We have designed and constructed a functional Cryostat at JPL specifically for this experiment, which includes a specially designed helium dewar and Cryo-Insert. The helium dewar was built in such a way that the superconducting magnet can be cooled continuously to its operating temperature of 2.2 Kelvin by a Lambda Plate, therefore allowing the magnet to operate without interruption even during liquid helium transfer. Much of the effort of this project was devoted to development and testing of the apparatus and technologies necessary for the eventual high-precision measurements. Examples include the brand-new cryo-insert, lambda plate, cryogenic high-field multi-layer magnetic shields, superconducting magnetic shields, and liquid helium sample cells. After completion of the construction, we have tested our magnetic shielding scheme and found out what need to be improved in order to achieve our shielding goals. Cryogenic high-field magnetic shielding has proven to be quite a challenge. The experimental space in our experiment, like those in most of other low temperature experiments, is limited, presenting quite a challenge to accommodate massive shields necessary for high-field shielding. We believe the way forward resides on multiple shields targeted at individual locations. During the magnetic shielding test, we also measured the field profiles of the magnet by using multiple low-temperature Hall probes. The results showed that the field profiles did not appear to be altered significantly by the magnetic shields. We also tested the Lambda plate with the magnet being fully energized at 114 Amperes. The inlet flow impedance, controlled by a newly-built manual cryogenic valve, and pumping capability were tested and adjusted under magnet's working conditions. Through this test, we proved the concept and feasibility of the lambda plate cooling adapted for our specific cryostat configuration and operation conditions.
Bibliography Type:	Description: (Last Updated: 04/09/2008)
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Articles in Peer-reviewed Journals	Chui T, Zhang B, Barmatz M, Hahn I, Penanen K, Hays C, Strayer D, Liu YM, Zhong F, Young J, Radey T, Jones J, Galitzki N, Li N, Lo LY, Horikoshi S, Hollen S, Paik HJ. "Cryogenics for lunar exploration." Cryogenics 2006 Feb-Mar;4 (2-3):74-81. http://dx.doi.org/10.1016/j.cryogenics.2005.10.006 , Feb-2006
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